

MINING INDUSTRY

BIG DATA GROUP PROJECT TEAM 2

RENZO GEVART - 02103651
DOUWE PAUWELS - 02105447
HENRI BOLLIOU - 02112399
EMILE SCHOEMANS -02006755
VINCENZO VANTORNOUT - 02108145



Agenda

01 Managerial overview

02 When does financing become likely?

03 How do announcements impact stock prices?

04 Technical overview

05 Pipeline and model discussion

06 Insights and technical challenges

07 Questions



Managerial Overview

Issues at hand

- Early-stage mining firms cannot fund growth internally and must repeatedly access external capital
- Financing announcements systematically destroy short-term shareholder value through dilution
- Markets do not fully price financing risk ex ante, despite observable warning signals
- This mispricing creates predictable risk and exploitable opportunity for investors

Evidence from prior research

- Mining stock prices respond strongly to company news and announcements even before relaese date.

The Market Response to Exploration, Resource and Reserve Announcements by Mining Companies: Australian Data
(Bird et al., 2010 – UTS Working Paper)

1. **Can news articles and stock information help predict financing announcements in mining firms?**
2. **What is the stock price impact once these financing announcements occur?**



When does financing become likely?

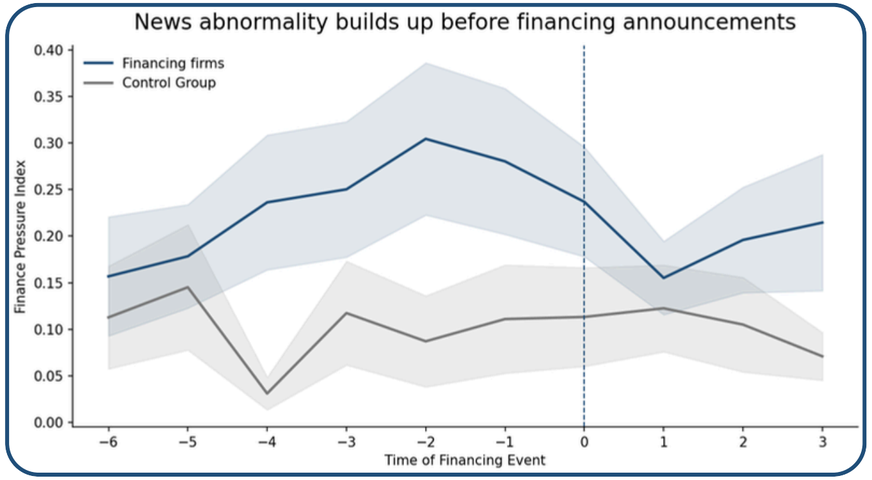
- NEWS & COMMUNITY SIGNALS AS EARLY WARNING
- STRONG PERFORMANCE CREATES THE WINDOW
- EXPLORATION RESULTS OPEN A SHORT FINANCING WINDOW
- MARKET STRESS AS CLEAR SIGNAL

FINANCING ANNOUNCEMENTS RARELY COME AS A SURPRISE, THEY CAN BE ANTICIPATED WHEN PERFORMANCE, MARKETS, AND INFORMATION SIGNALS ALIGN.

News & community signals as early warning

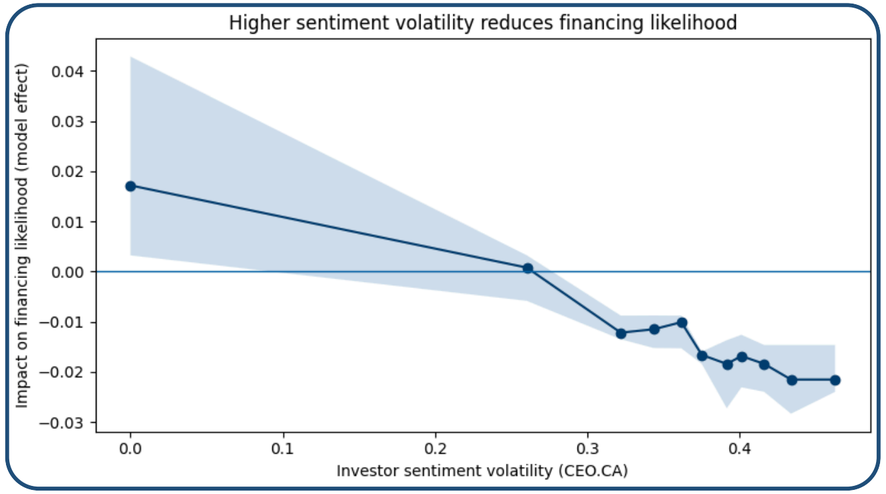
STRATEGIC COMMUNICATION IS A LEADING INDICATOR

- Shift in communication behaviour of the company in news articles strongly precedes financing



SENTIMENT STABILITY MATTERS MORE THAN BUZZ

- Stable investor sentiment (on CEO.CA) is a signal for high financing likelihood
- While volatile sentiment is a sign of financing announcements, especially for risky firms



ATTENTION QUALITY > ATTENTION QUANTITY

- Early, focused attention is informative
- Sustained or extreme chatter signals noise, speculation, or controversy

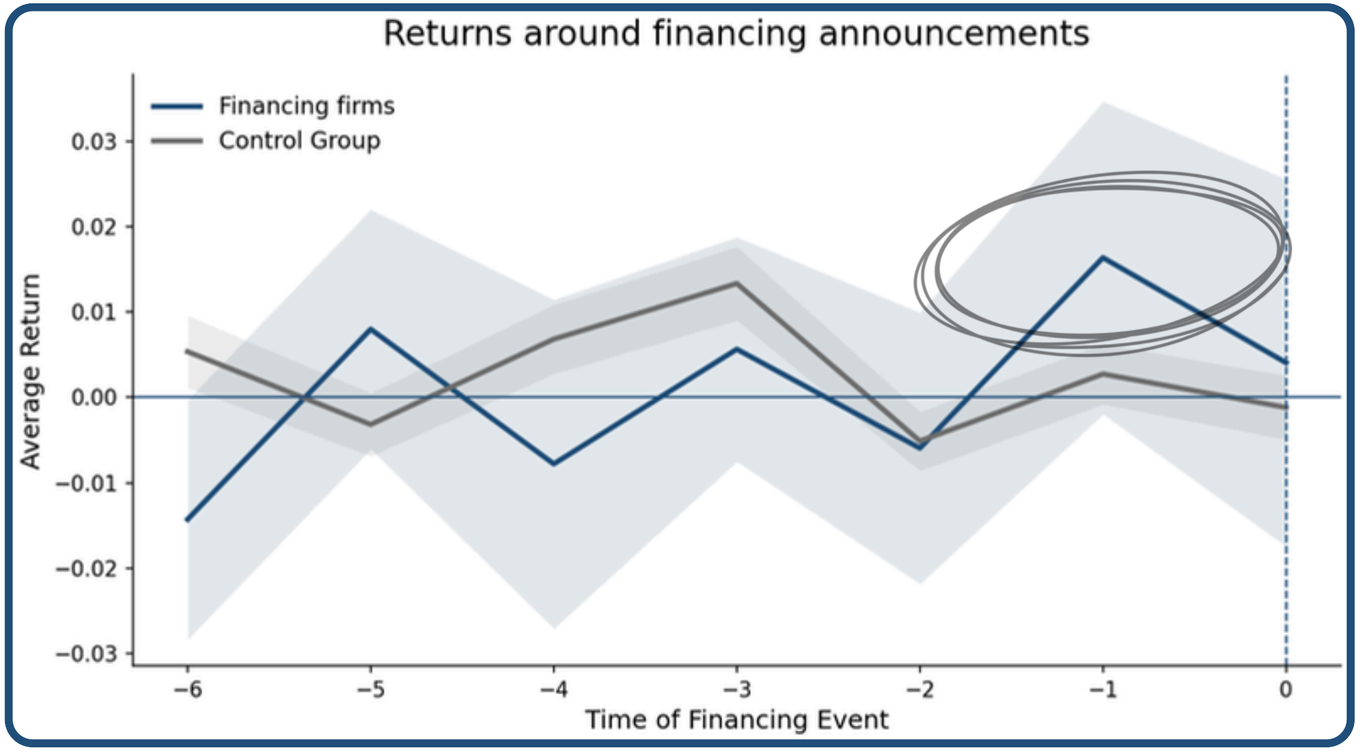
Successful financing requires expectation management, not hype

Positive performance creates the window

FINANCING IS MORE LIKELY AFTER RECENT POSITIVE RETURNS

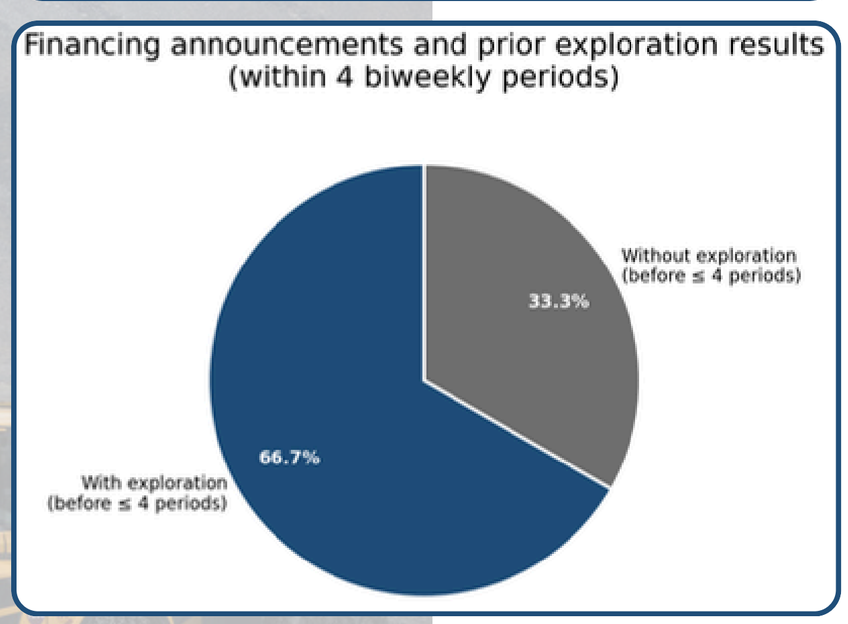
FIRMS TEND TO AVOID RAISING CAPITAL AFTER WEAK OR NEGATIVE PERFORMANCE

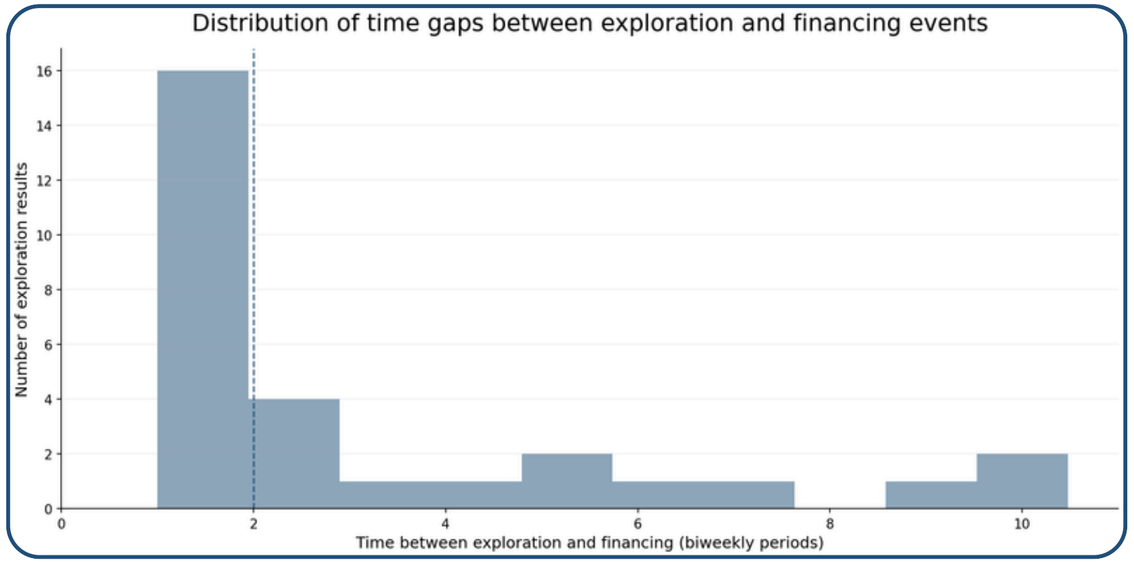




Recent Exploration results as trigger

~2/3 OF FINANCING ANNOUNCEMENTS FOLLOW EXPLORATION RESULTS WITHIN 4 BIWEEKLY PERIODS





FINANCING TYPICALLY OCCURS WITHIN 1–2 BIWEEKLY PERIODS AFTER EXPLORATION UPDATES

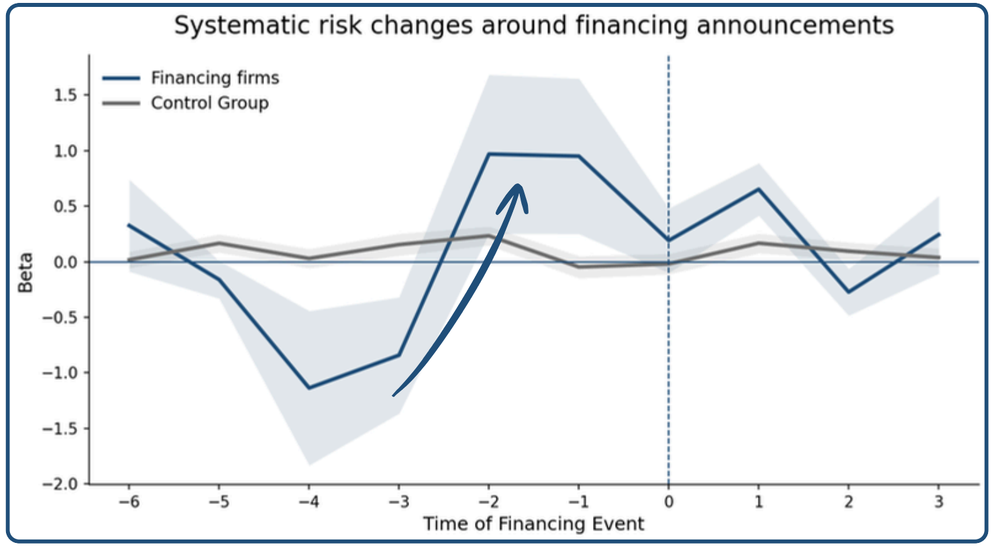
Market stress as clear signal

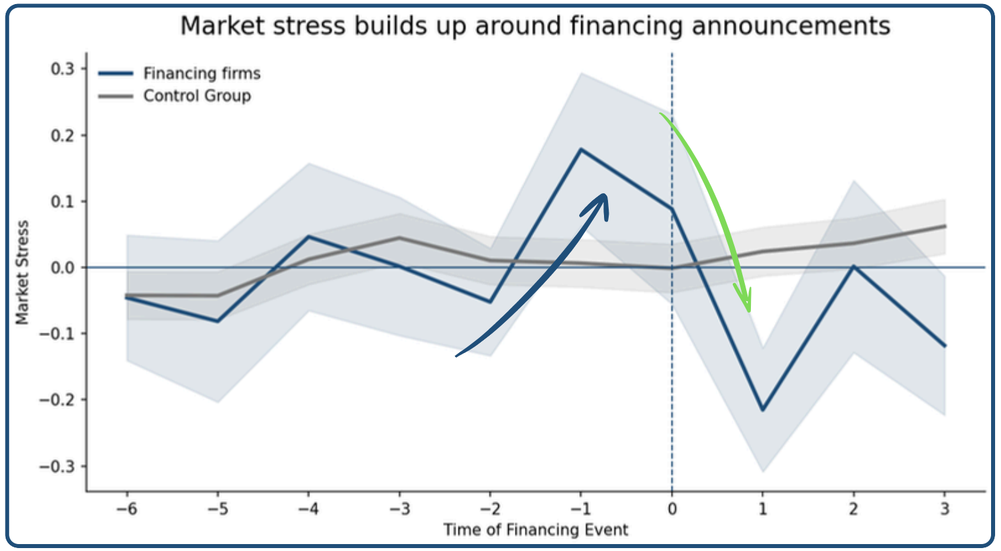
- **Low Return Volatility → Low Financing Likelihood**
- **High Return Volatility → High Financing Likelihood**

Before financing announcements, firms that raise capital show temporarily elevated beta

Market stress captures how unstable and reactive a stock's price behavior is

- It captures volatility, gap risk, and intraperiod instability
- It builds up ahead of financing announcements and declines afterward

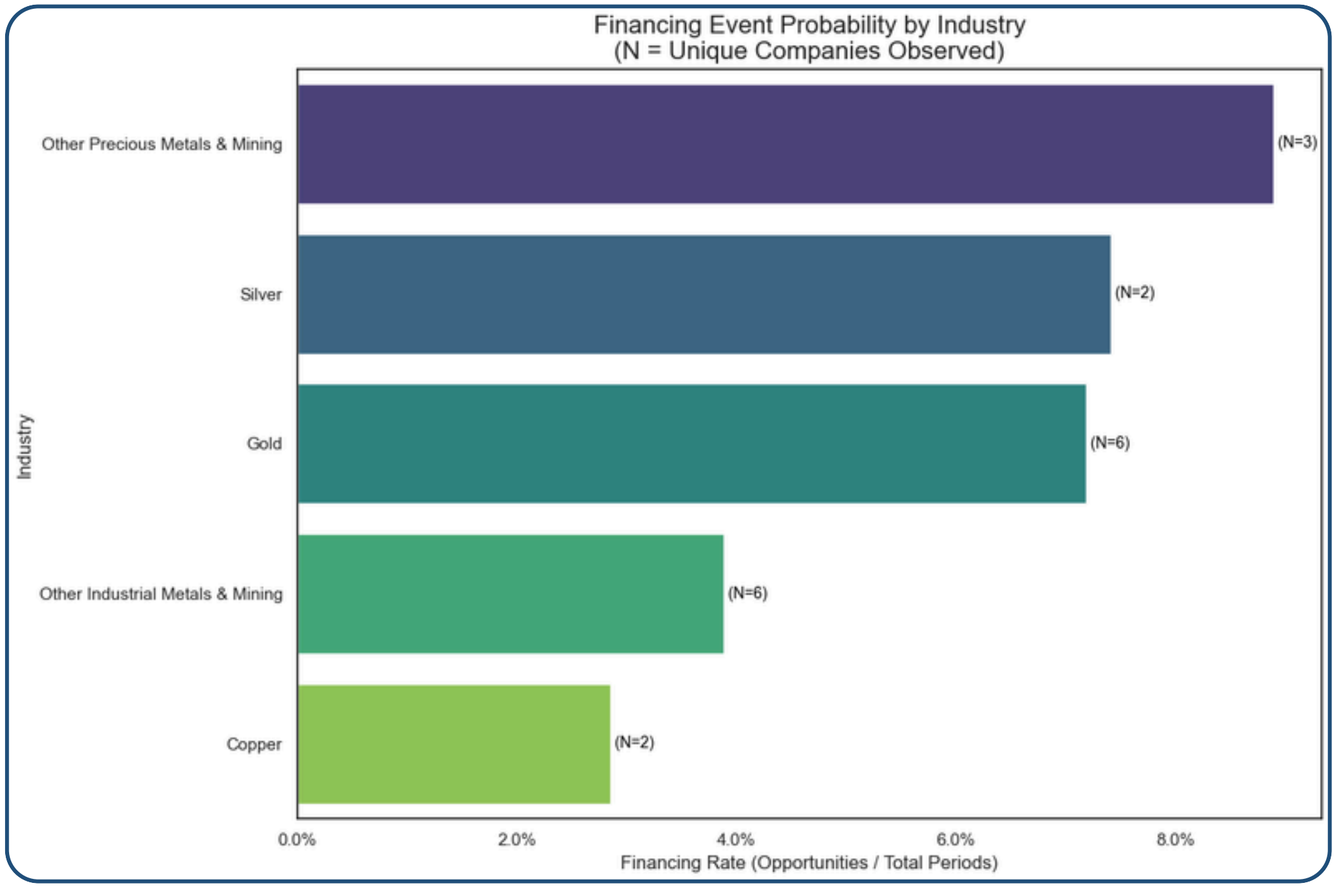






Comparison Across Industry

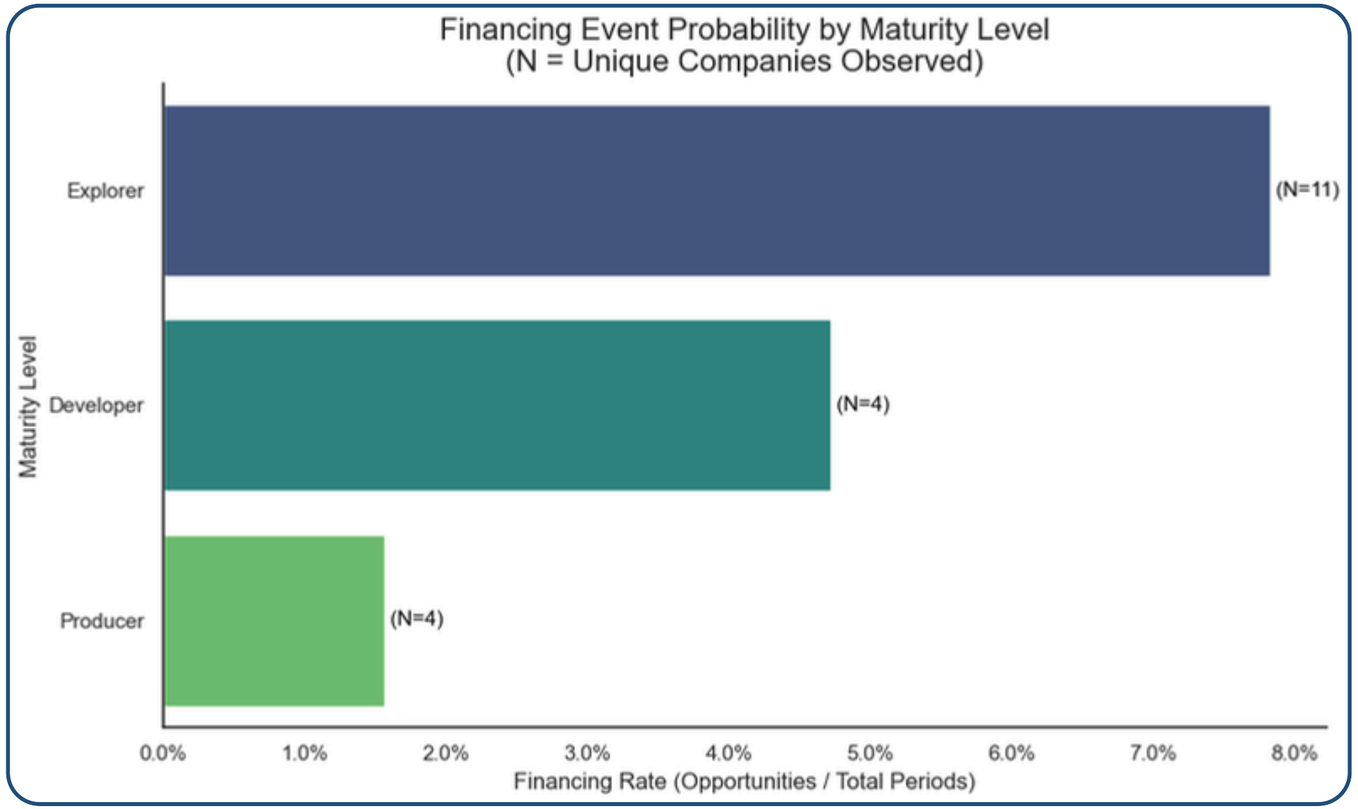
WHAT IS THE FINANCING ANNOUNCEMENT PROBABILITY BY INDUSTRY?





Comparison Across Maturity Level

WHAT IS THE FINANCING ANNOUNCEMENT PROBABILITY BY MATURITY LEVEL?





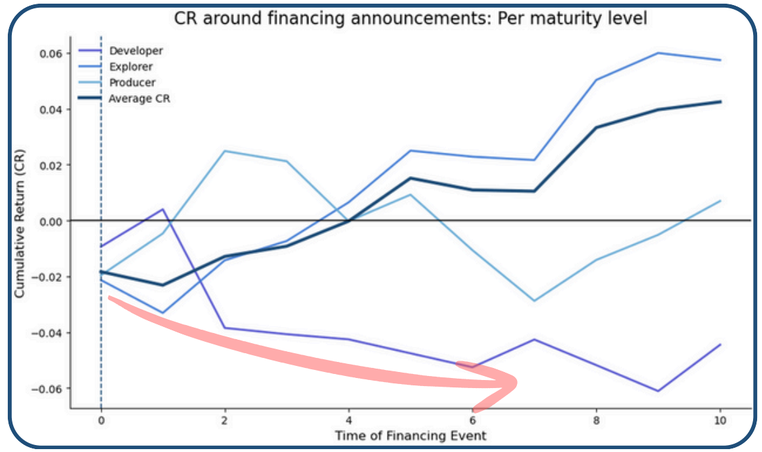
Impact of financing announcements on stock price

IMMEDIATE MARKET IMPACT

- Stock returns decline significantly on the announcement day ($p < 0.05$)
- Investors interpret financing as dilutive or risk-increasing

DOWNSIDE RISK DOMINATES

- Developer-stage firms experience persistent negative CRs
- Explorers and producers show neutral to mildly positive reactions





What do we offer investors?

The model is designed to manage dilution risk, not to optimize trading precision.

Why accuracy is not the right metric

- Financing events are rare but **costly**
- **Missing** a financing event is far more expensive than reacting to a false warnings
- Risk management therefore **prioritizes early detection** over exact timing

How the model should be interpreted

An early-warning system that flags periods of elevated financing risk before announcements occur.

Input used

Market behavior, risk exposure, news activity, and company communication



How does this change investor decisions?

Model outcomes (simple framework)

Early warning hits → financing risk flagged before announcement
Missed events → financing without prior warning
False alarms → elevated risk without subsequent financing

Why this matters economically

Early warning hits are most valuable
→ allow exposure reduction, position sizing, or delayed entry
The model identifies ~76% of financing events in advance
→ substantially reducing unhedged dilution risk

False alarms (important reframing)

- Economically less damaging than **missed events**
- Preferable in a setting where financing announcements have **consistently negative price effects**



Technical Overview

- Data & pipeline
- Modeling approach
- Model interpretation
- Technical challenges

From raw data to interpretable early-warning signals



Data & Pipeline

1) DATA COLLECTION

From multi-source datasets including mainstream news, specialized financial publications, and investor sentiment forums

2) DATA CLEANING

- Standard preprocessing for numerical market data
- Categorical encoding (industry, maturity, region)
- Text cleaning and normalization for news and forum data

3) FEATURE ENGINEERING

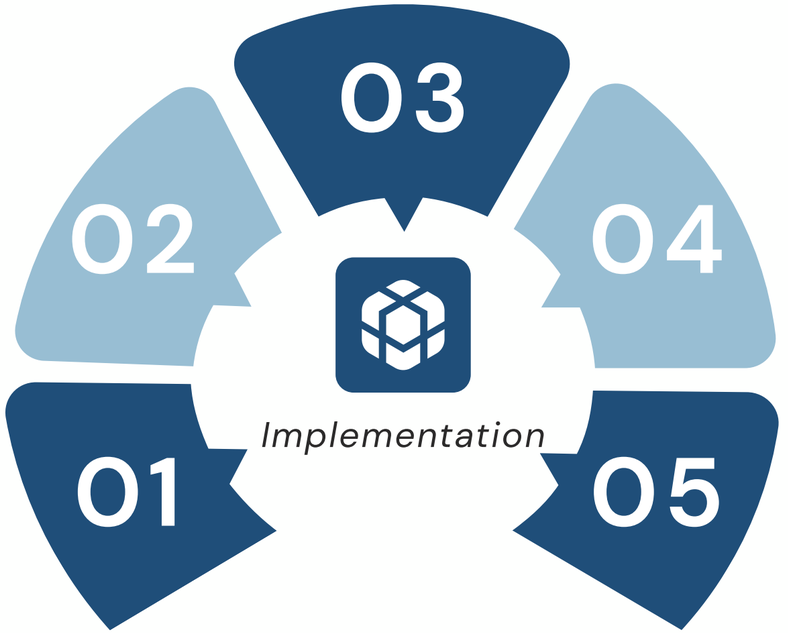
- Market features
- Sentiment analysis on news and investor communication
- Sentiment level and sentiment volatility measures

4) TEMPORAL AGGREGATION

- Aggregation of daily and event-level data into biweekly periods
- Creation of a firm-period structure

5) FEATURE SELECTION & INTEGRATION

- Merging all features into a single biweekly base table
- Redundancy reduction using correlation analysis



01 Data Pipeline

Data Scraping

Ticker specific sources:

- **Yahoo Finance**
- **Company news pages**
- **CEO.CA (investor forum)**

Beta creation

Industry specific sources:

- **De Tijd articles**
- **Financial Times articles**
  - Keyword filtering + allowed for some overlap

DE TIJD

CHESAPEAKE GOLD CORP.

yahoo!finance

CEO.CA

FT FINANCIAL TIMES

GOLD & SILVER

FIRST ANDES SILVER

02 Data Pipeline

02 Data Cleaning

General

Validity & consistency checks
Handling of missing data

Normalization & scaling
- Stock market features are standardized relative to each firm's own history
- Sentiment features are standardized globally

One-hot encoding of categorical features

Textual data

De-duplication

Noise removal & Standardization
- Lowercasing
- Remove URLs
- Removal of punctuation, tickers and numbers/values
- Removal of context words
- etc.

Forum vs article data

Temporal alignment
- Text item are timestamped and aligned to a biweekly period
- Only information available before the financing window was included

Feature creation
- Lemmatization
- Tokenization

03 Data Pipeline

Feature Engineering

Feature creation blocks:

- Stock data and derived features
- CEO.CA sentiment features
- Company sentiment features
- Industry sentiment features

- Yahoo Articles: FinBERT & Ollama
- Company new pages: FinBERT & Ollama
- De Tijd: FinBERT & Ollama
- Financial Times: Finbert & Ollama
- CEO.CA: VADER & RoBERTa

Key risk & signaling features

- Biweekly Beta (industry-relative risk exposure)
- Financing Pressure Index

FinBERT

RoBERTa

TextBlob

Ollama

03 Data Pipeline

Feature Engineering : Financing Pressure Index (FPI)



FinBERT embeddings

Dimensionality Reduction
UMAP

Clustering
DBSCAN

FPI
Cosine Distance of FinBERT embedding to the baseline cluster embedding

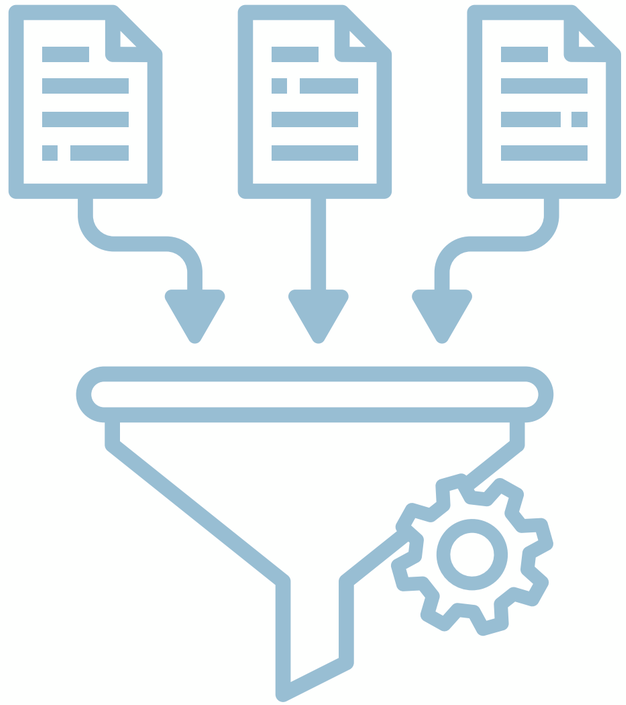
04

Data Pipeline

Temporal Aggregation

What is aggregated

- Daily market features (returns, volatility, beta, stress)
- Event-level news and sentiment signals
- Forum activity and sentiment



How aggregation is performed

- Signals are aggregated per ticker × biweekly period
- Industry-level signals are merged using industry × period
- Only information available before the period end is included

05 Data Pipeline

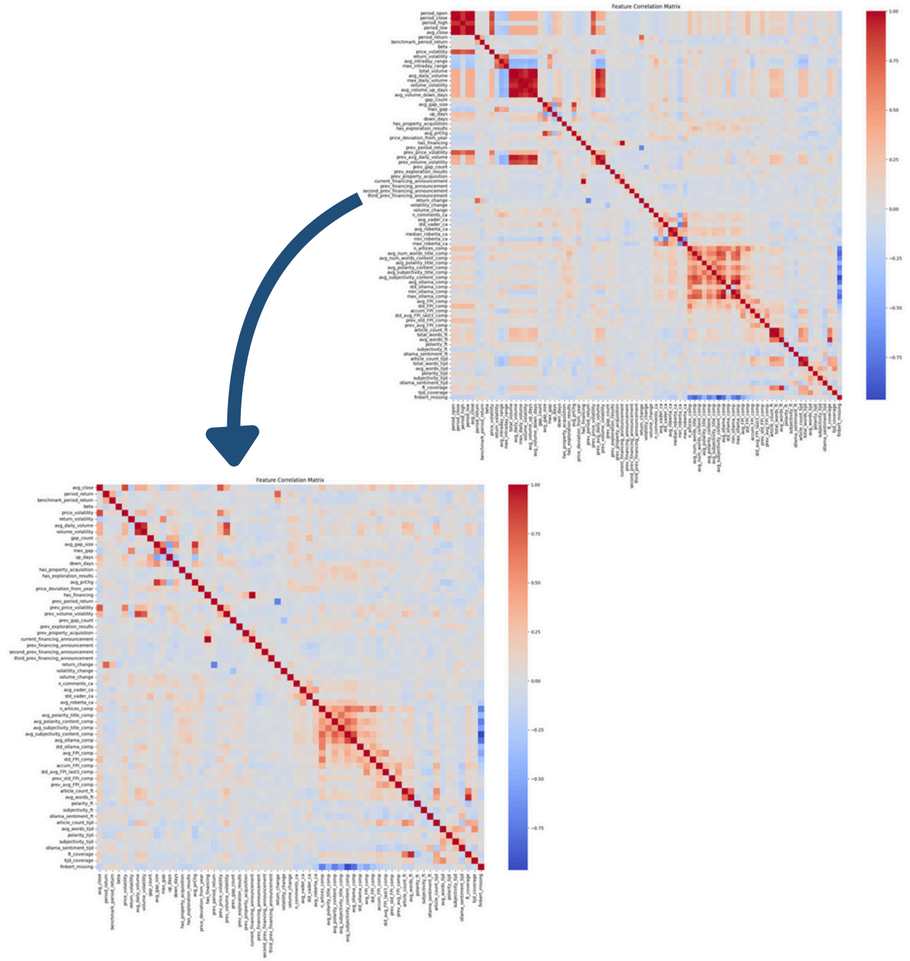
Feature Selection

Objective

Removing highly correlated signals within feature blocks, while maintaining diversity across blocks to avoid overfitting.

Method

- Correlation matrix used to identify redundant features
- Redundant feature pairs removed, most interpretable features retained



Simplify feature blocks without collapsing them



Modeling approach

Baseline Model

- Input: stock data features
- Logistic Regression

Extended Models

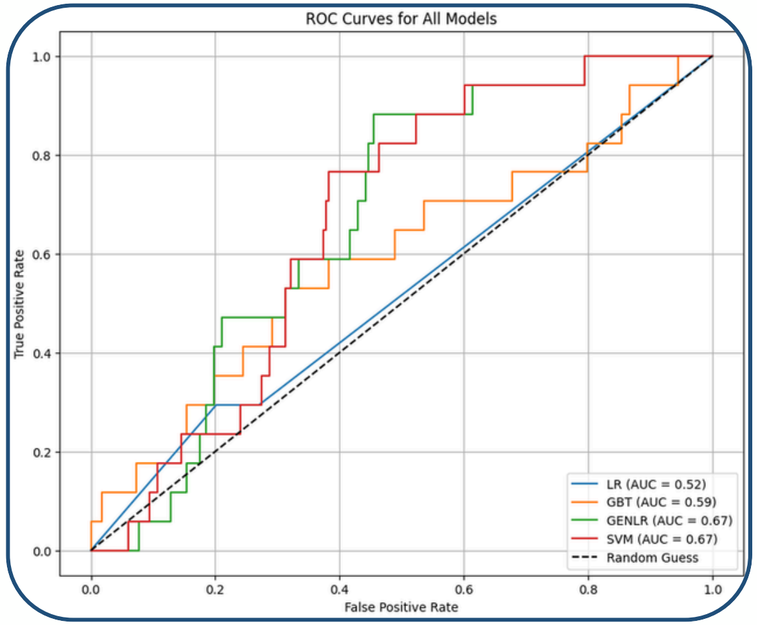
- Input: stock data + sentiment data features (reduced feature set)
- Logistic Regression, with Elastic Net Reguralization
- Gradient Boosting TC
- SVM

THRESHOLD TUNING FOR IMPROVING RECALL



Model Interpretations

AUC



Best Model: Support Vector Machine

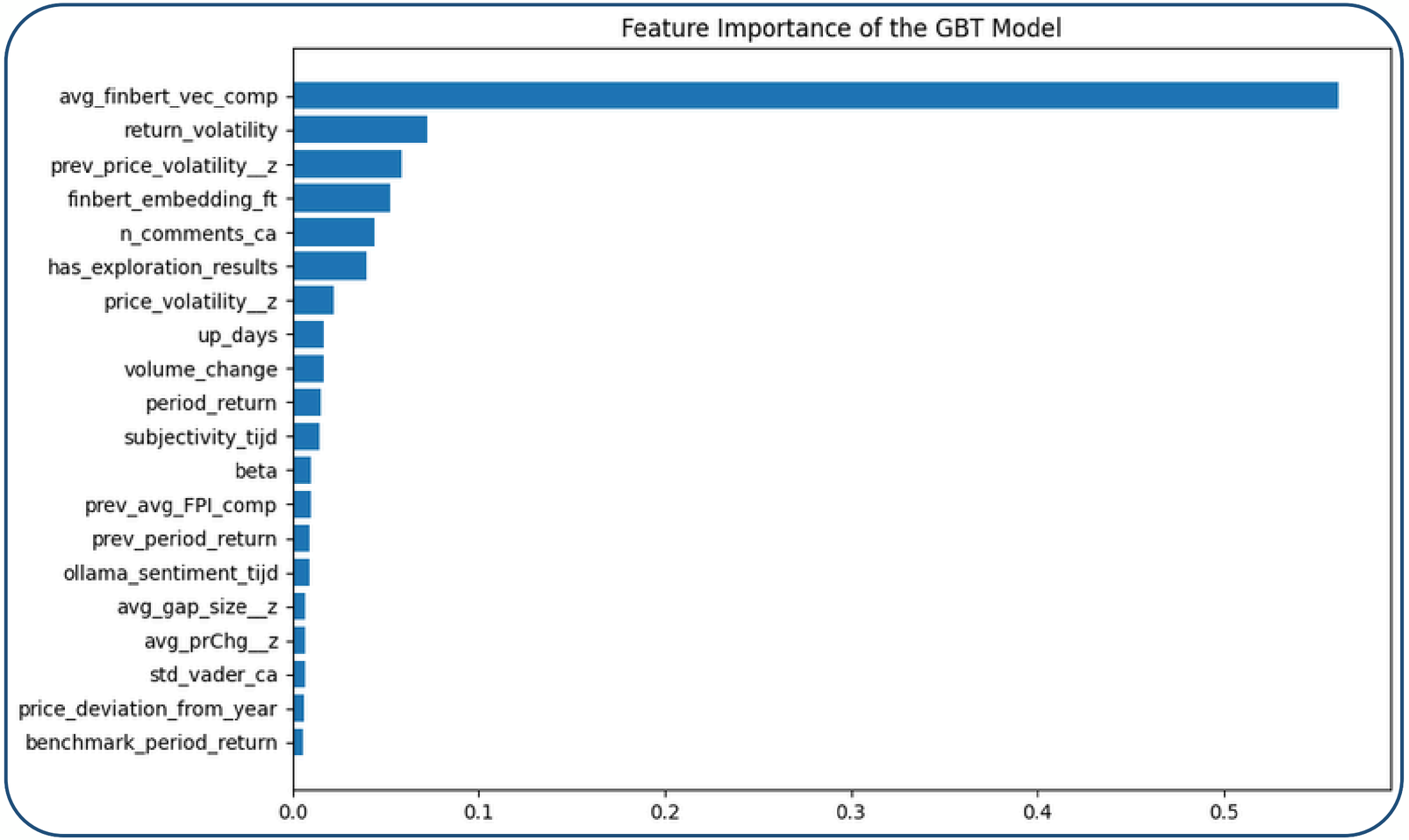
- Accuracy: 0.61
- Precision: 0.12
- Recall: 0.76
- F1 - Measure: 0.21

Feature importance: Gradient-Boosted Tree Classifier

Dominant Feature Groups

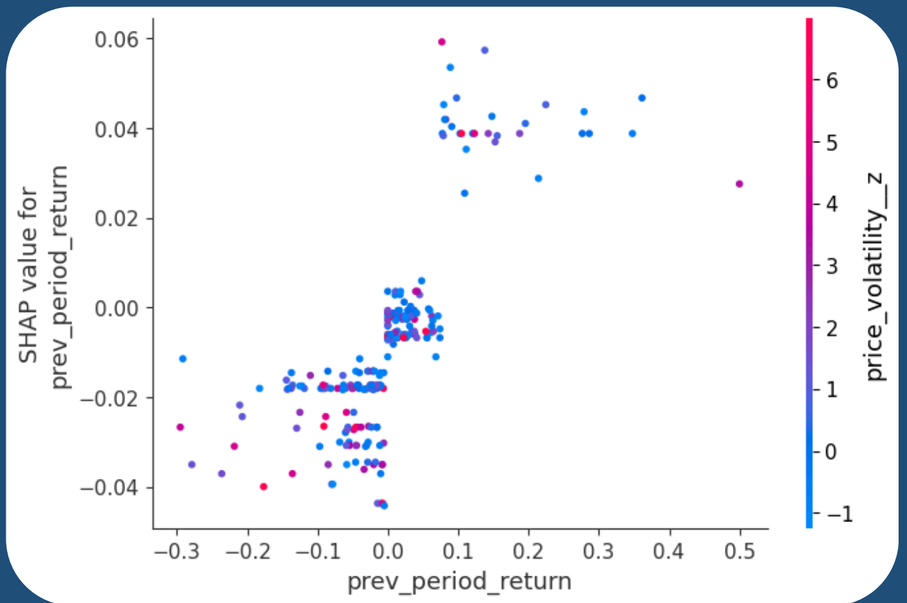
- Company-level sentiment embeddings
- Volatility-related features
- Communication activity proxies (FPI, comment volume)
- Systematic risk (beta)
- Event indicators

High-dimensional sentiment embeddings and volatility features dominate model splits.

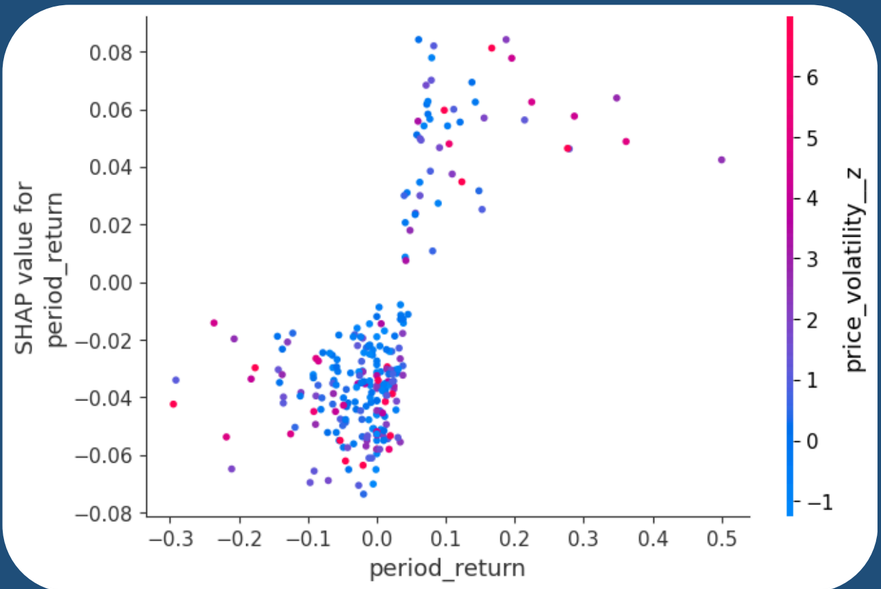


Shap Dependence Plots: Market Performance

Number of comments in period i

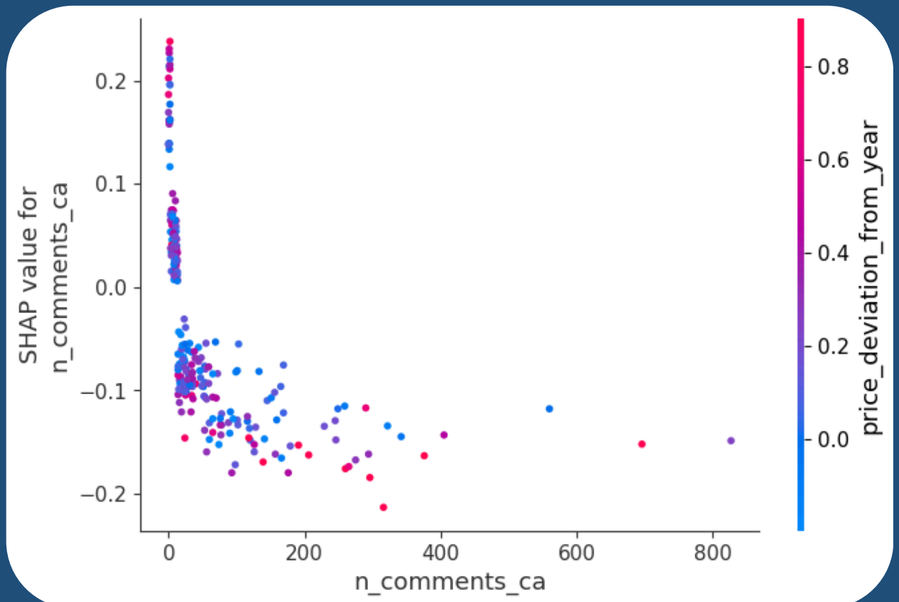


Return over current biweekly period

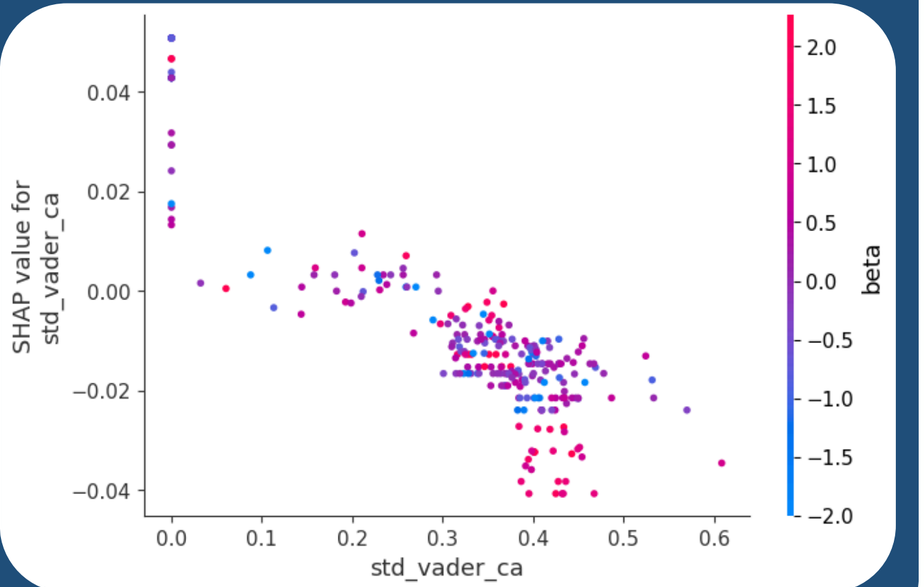


Shap Dependence Plots: CEO.CA Insights

Return over previous biweekly period i



Stdv of vader sentiment in period i





Technical Challenges

- CLASS IMBALANCE
- SMALL AMOUNT OF DATA
- TRADE-OFF: RECALL AND EARLY DETECTION
- INTERPRETABILITY VS PREDICTABILITY



THANK YOU

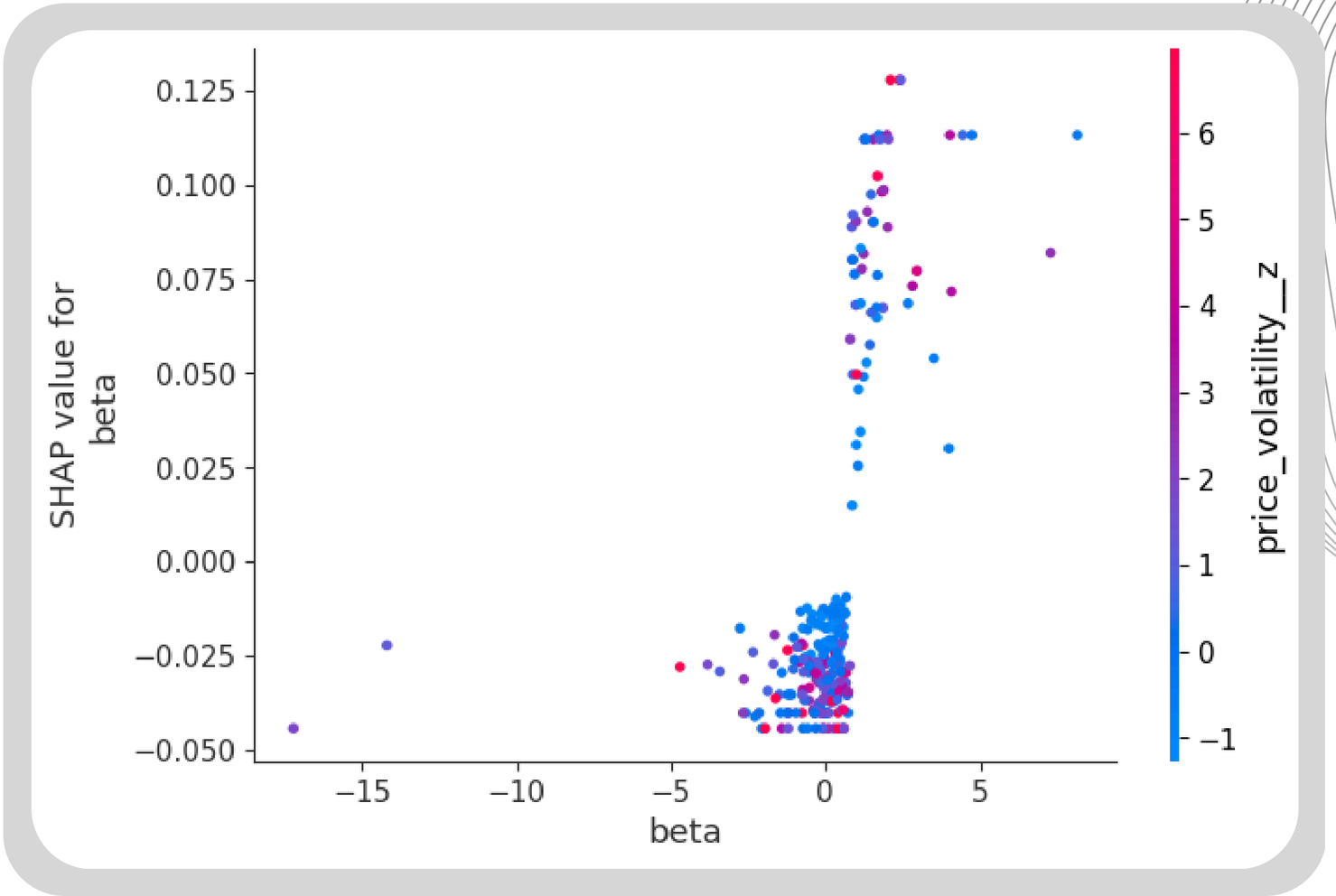
Any Questions?

Full Text Cleaning for News Articles

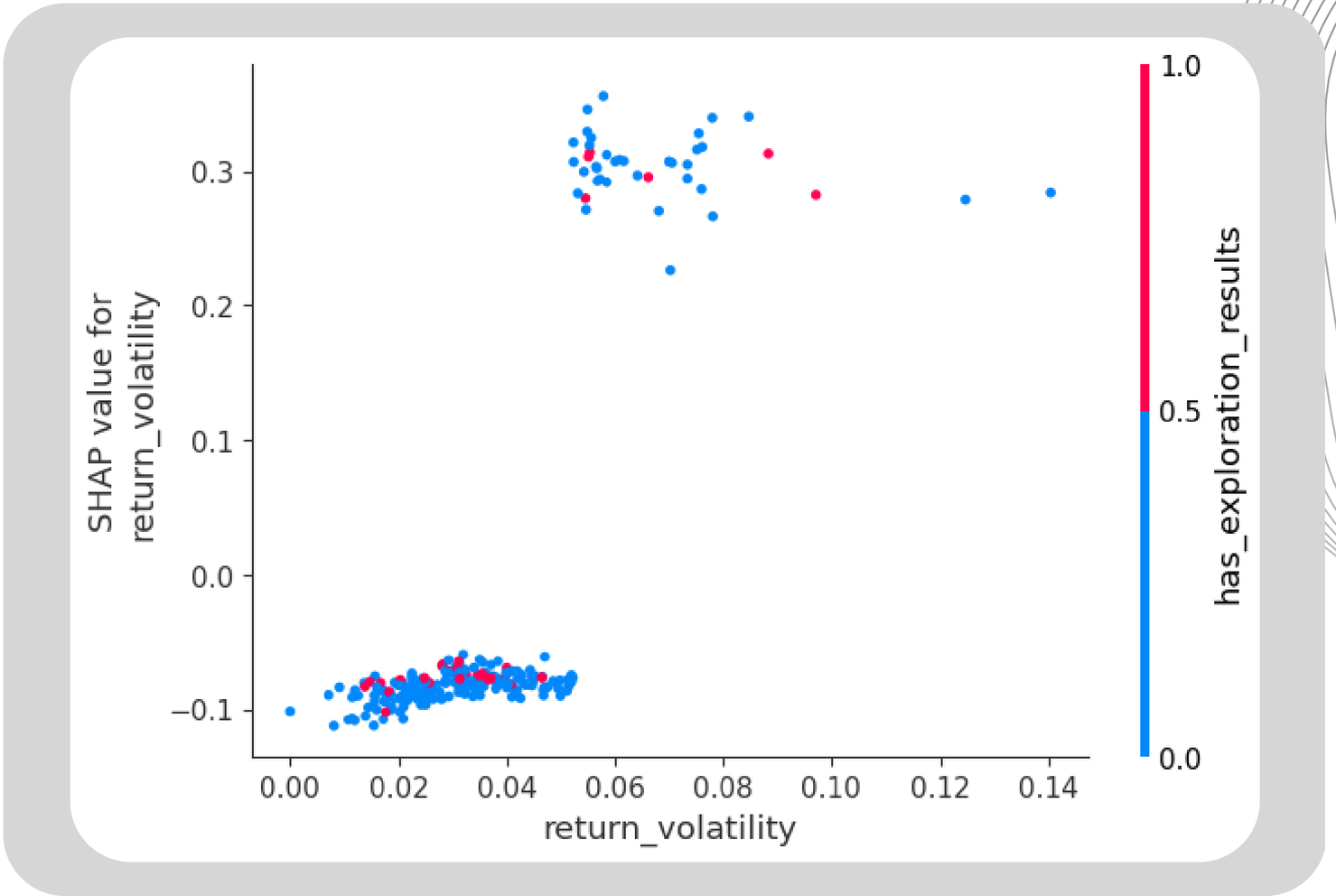
- Lower Casing text
- Removal of missing or empty text entries
- Removal of Punctuations
- Removal Of Numbers
- Removal Of Special Characters
- Whitespace Normalization
- Tokenization
- Stopword Removal
- Lemmatization
- Removal Of tickers
- Domain-Specific Removal of text (e.g, For Further information, company information: ..., etc.)

FEATURE IMPORTANCE & SHAP

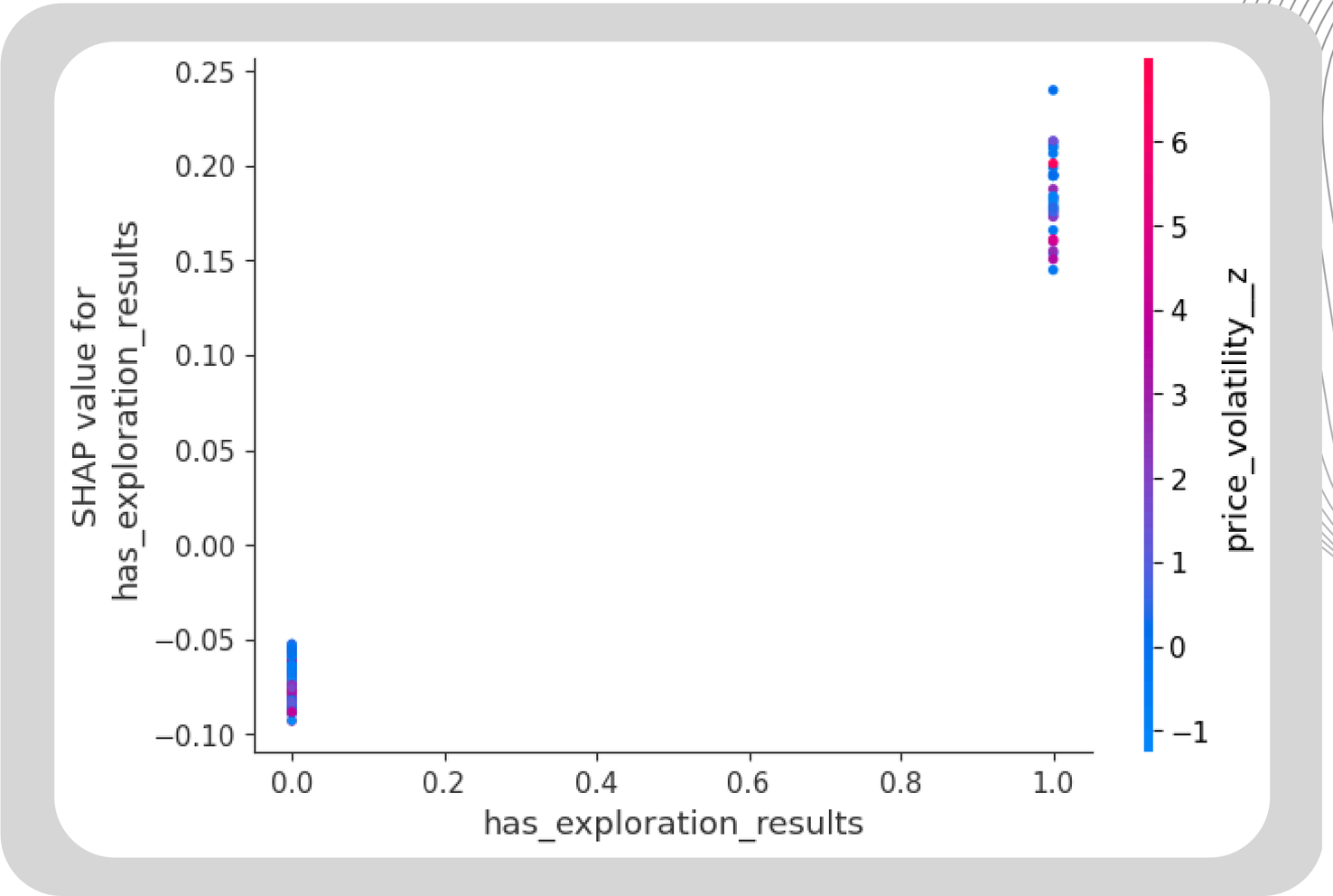
Shap Dependence Plots: Risk Profile (Beta)



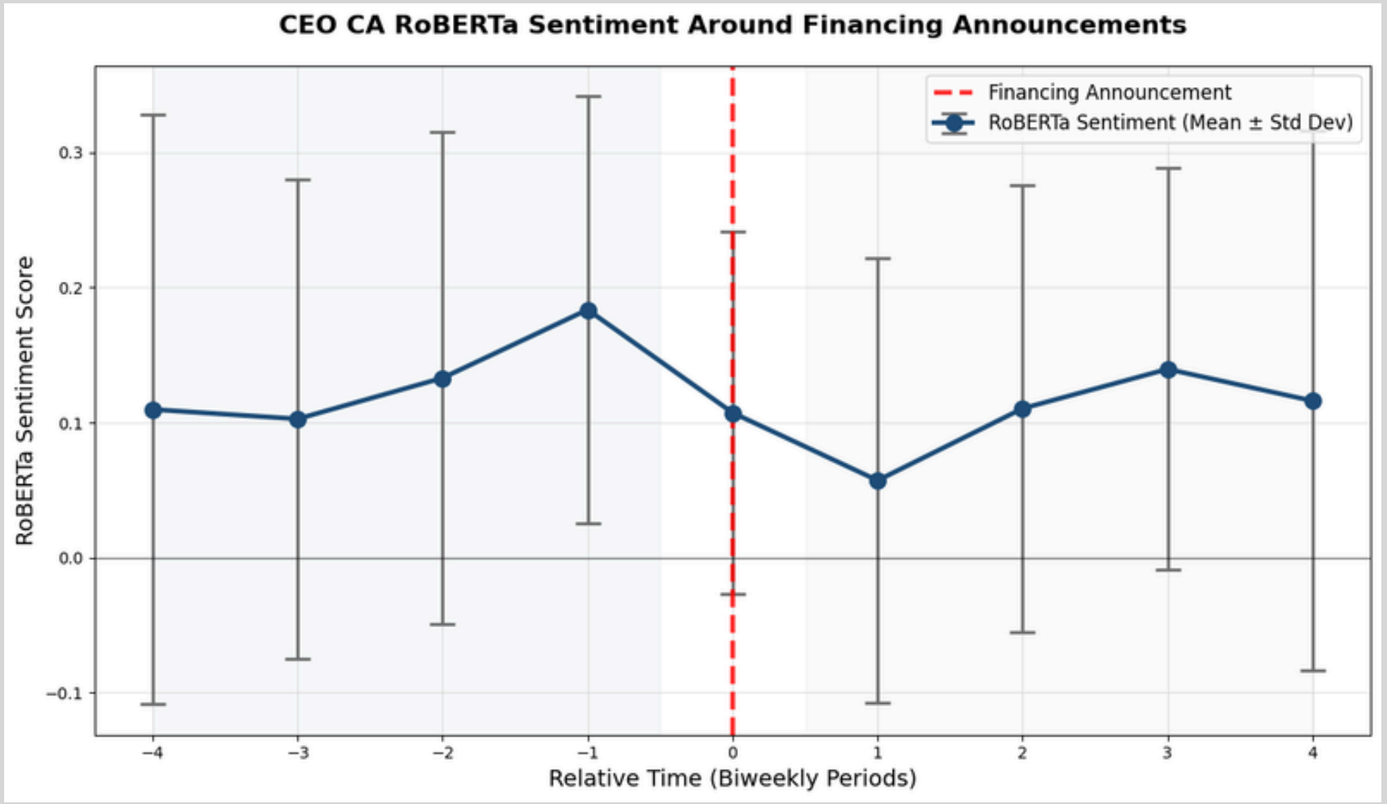
Shap Dependence Plots: Return Volatility



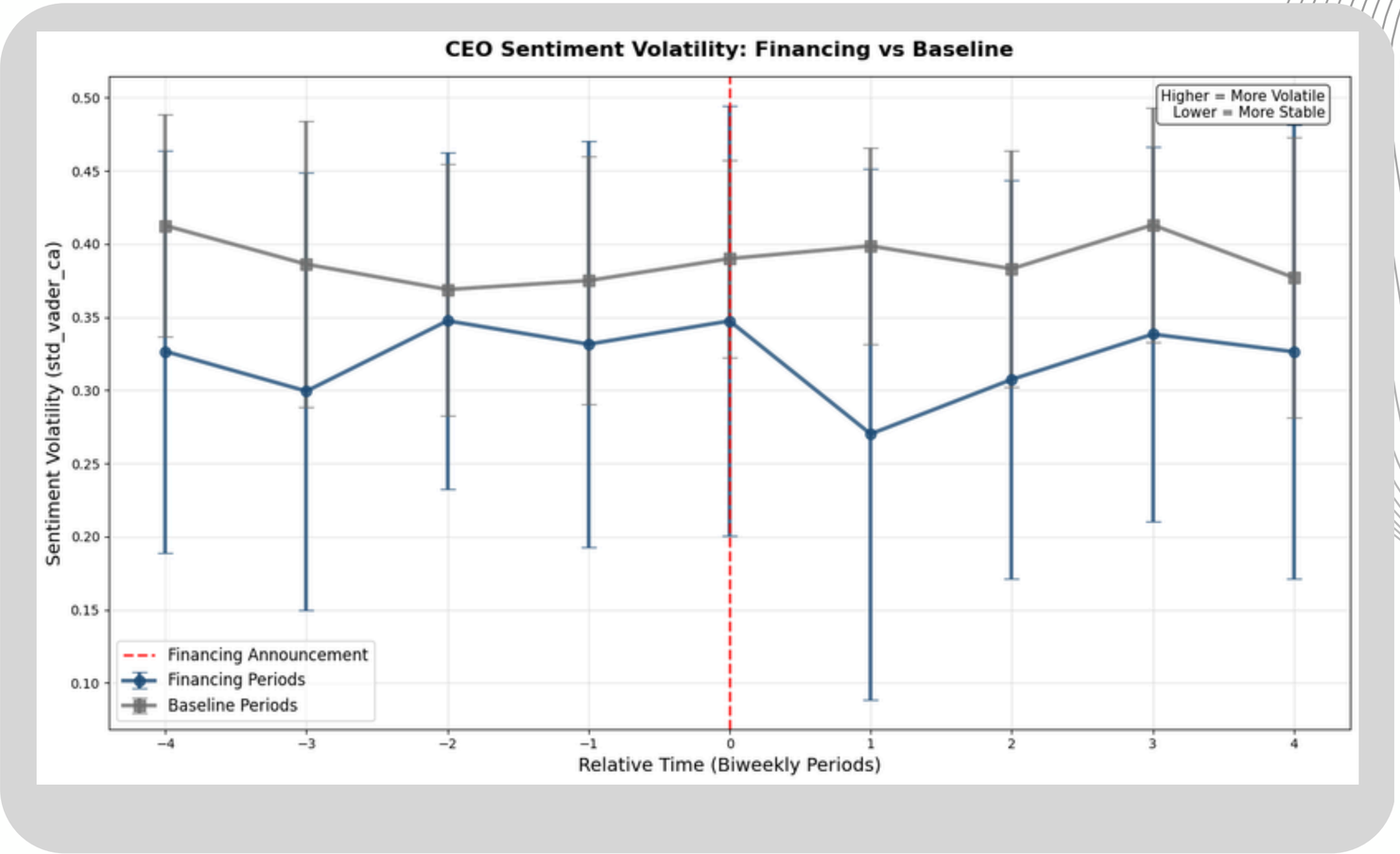
Shap Dependence Plots: Exploration Results



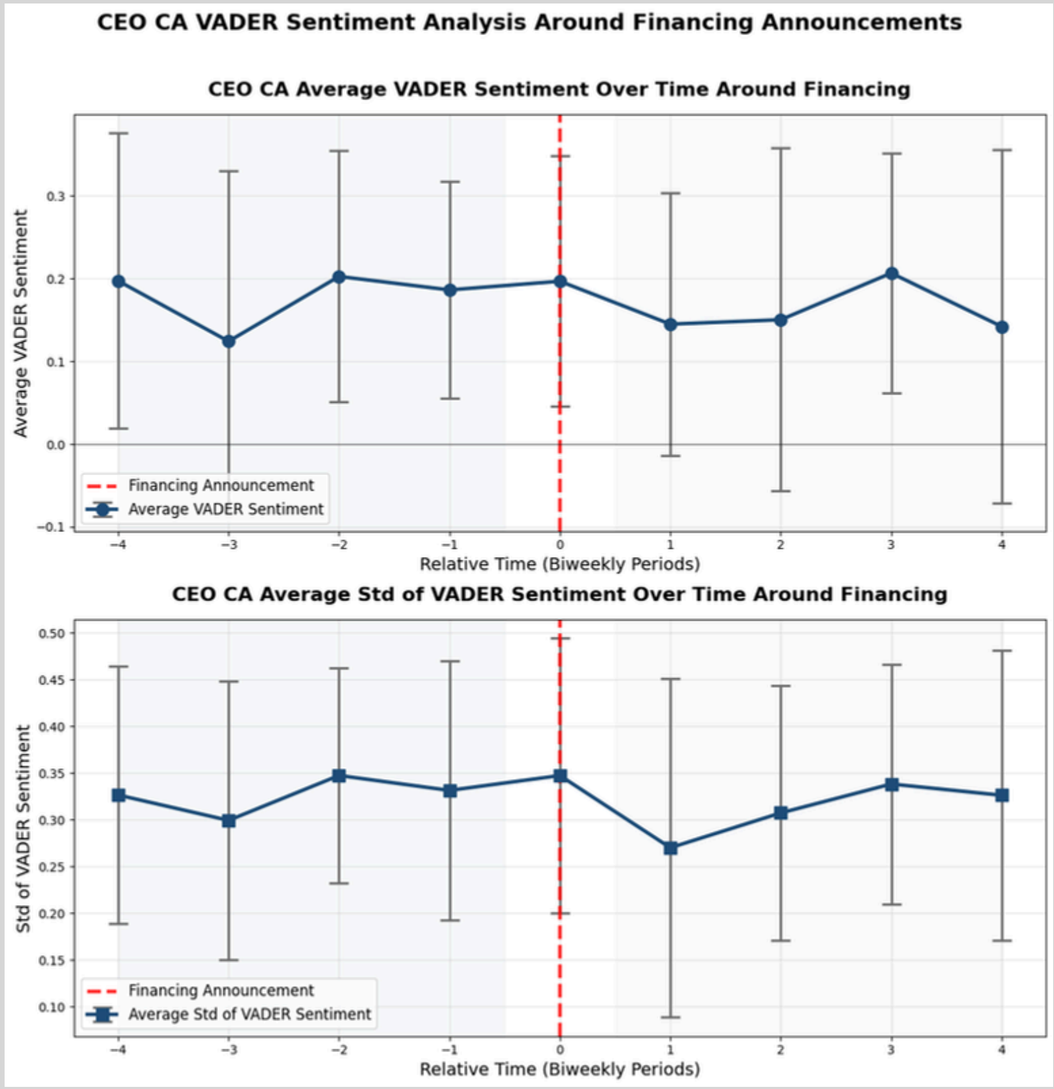
RoBERTa Sentiment Around Financing



CEO Sentiment Volatility Financing vs Baseline



CEO Sentiment Volatility Financing vs Baseline



Keyword filtering + allowed for some overlap

WHY

Solving the Overlap Problem

- The Issue: Financial articles often mention multiple commodities (e.g., "Gold vs. Copper").
- The Goal: Isolate pure sector signals and prevent general "mining noise" from diluting specific results.

HOW

Hierarchical Regex & Transformation

- Priority Hierarchy:
  - Independent: Gold, Silver, Copper (High-priority).
  - Conditional: Specialized metals assigned only if T1 is absent.
  - Residual: General mining keywords assigned only if no specific metals are found.
- Robust Matching: Used word boundaries (\b) and dual-field scanning (Title + Content) to maximize precision.
- Pipeline Logic: Implemented a Wide-to-Long transformation using F.explode() to handle articles with multiple labels.

RESULT

Multi-Sector Benchmarking

- Granularity: Each article is correctly mapped to its specific peer group(s).
- Leverage: One high-quality news item can now inform the averages of multiple industries simultaneously, ensuring no data is wasted.
- Integration: Created a clean industry column ready for bi-weekly PySpark aggregation.

Financing rate used in slide 9 & 10

$$\text{Rate} = \frac{\sum \text{Financing Events}}{\sum \text{Observed Bi-weeks}}$$

Problem:

- **Bias Mitigation: Simple event counts favor sectors with more data/companies.**

Shifted focus to "Periods of Exposure" (total time monitored).

Aggregation Strategy:

Leveraged PySpark's .agg() and F.count(F.lit(1)) to calculate the total "observation opportunities" for every sector and maturity level.

=> every 2-week block is a "chance" for a company to raise money

Benchmark: Establishes the "Hazard Rate" (the prior probability or "baseline" for the model).

Definition:

empirical probability of a financing announcement occurring within a 14-day window.

Focus:

It shifts the perspective from a simple "count of events" to a normalized risk signal per period of exposure.